



Society of Petroleum Engineers

SPE-229131-MS

Mitigating the Impact of Wind Sensor Failures in Drone-Based Methane Mass Flow Measurements: Lessons from the TADI Blind Release Campaign

T. Bullock, J. Kesti, and M. Irjala, Aeromon Oy, Helsinki, Finland

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229131-MS](https://doi.org/10.2118/229131-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

In June 2024, Aeromon participated in a controlled methane release test campaign organized by the International Methane Emissions Observatory (IMEO) and Stanford University at the TADI test centre in Lacq, France. During the comparison between reported and actual methane release rates, Aeromon's measurements were found to be consistently lower than the known values. A detailed investigation traced the issue to an unexpected malfunction in the on-board wind sensor. To fully understand the impact, Aeromon reviewed its entire measurement process—from data collection to final analysis—and identified areas where quality assurance could be improved. This paper presents the main steps and findings from that review. To independently assess the reliability of Aeromon's method, a re-analysis was conducted using wind LIDAR data provided by TotalEnergies. This analysis showed a much stronger correlation with the actual release rates, improving the linear fit from a slope of 0.47 to 0.92. The results confirmed that Aeromon's system performs accurately when supported by correct wind data. Importantly, the analysis also demonstrated that Aeromon's measurement system can effectively use wind LIDAR data, something not previously tested. Integrating LIDAR as an alternative wind source marks a meaningful step forward in improving airborne methane measurement and opens up new possibilities for real-world monitoring. The TADI campaign was a valuable research and development opportunity that led to important improvements in Aeromon's quality control and methodology. The lessons learned will enhance future measurement campaigns and contribute to progress across the industry in methane detection and reporting. By addressing challenges head-on, Aeromon continues to drive innovation in airborne emission monitoring, raising the bar for accuracy, reliability, and scientific integrity.

Keywords: Blind controlled test release, drone-based methane emission monitoring, site level quantification, quality control, quality assurance

Introduction

About Aeromon

Aeromon provides mobile emission monitoring services globally using its patented measurement devices and advanced analysis & reporting tools. The company's multi-parameter monitoring system platform is

primarily used for methane detection within O&G industries but is also capable of simultaneously measuring other gases such as CO₂, CO, H₂S, SO₂ and Non-Methane Volatile Organic Compounds (NMVOCs). As an ISO 17025:2017 accredited test laboratory ([FINAS 2024](#)) Aeromon places high importance on regularly participating in research and development-focused measurement campaigns. The normal setup used for data collection with a drone for mass flow rate analysis is shown in [Figure 1](#).

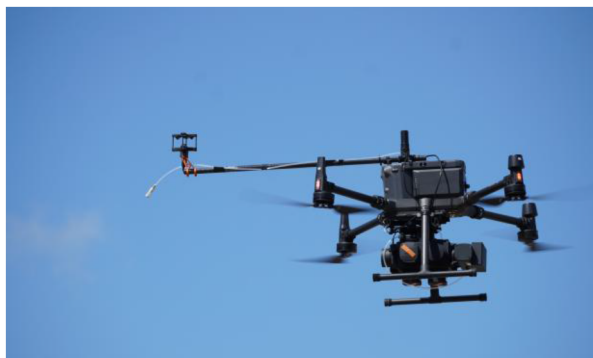


Figure 1—Aeromon's on-board measurement setup for mass flow rate measurements on-board a typical commercial drone. Sample intake and on-board wind sensors are located at the end of the probe, which is kept towards prevailing wind direction. The measurement device with gas sensors is located under the drone.

Research campaign and identification of faulty wind sensor

Aeromon participated in a test campaign designed to enhance understanding of current methane emission monitoring technologies through a blind controlled release setup. Over the past several years, Aeromon has taken part in more than ten such campaigns worldwide, all focused on assessing the effectiveness of different methane mass flow rate quantification methods.

The first week of this particular campaign was held from June 17th to 21st, 2024, at the TADI test centre in Lacq, France. It was organized by the International Methane Emissions Observatory (IMEO) under the UN Environment Programme, in collaboration with Stanford University. During the campaign, various controlled methane flows were released, and participating teams were tasked with independently estimating the release rates without prior knowledge of the actual values, enabling a quantitative assessment of each technology's performance. Aeromon took part throughout the entire first week, using the mass balance (MB) method to conduct its mass flow rate measurements.

After submitting its initial results to campaign organizers, the actual release rates were disclosed. A clear discrepancy emerged between the true values and Aeromon's estimates, even after accounting for expected variations due to weather conditions. This outcome did not match the outcome from previous comparison studies and prompted Aeromon's data analysis team to initiate a thorough investigation, which identified a malfunction in the on-board wind sensor as the primary source of error ([Hamedani Raja et al. 2025](#)). The review aimed not only to pinpoint the root cause of the deviation but also to verify the reliability and consistency of Aeromon's methodology against findings from previous controlled release studies.

Re-analysis of the mass flow rates based on wind LIDAR data

To better understand the prevailing weather conditions and establish a consistent point of comparison, mass flow estimates were recalculated using wind LIDAR data. While these LIDAR-based results serve as a validation tool, they are not intended to replace Aeromon's standard in-field approach, which relies on on-board wind measurements. However, this exercise offers valuable insights into potential future improvements. Since Aeromon's methodology has not yet been specifically tailored or validated for use with wind LIDAR data, further refinement could enhance its accuracy. For instance, correcting spatial mismatches between the locations of wind LIDAR measurements and methane concentration readings may lead to more precise results.

Aeromon's original flow rate estimates, previously deemed suboptimal due to wind conditions falling outside of the method's validated boundary conditions, are shown in Figure 2a. Corresponding estimates based on wind LIDAR data are presented in Figure 2b. Each figure includes a linear fit of the results. As each data point reflects a repetition of downwind wall measurements, the associated error bars incorporate both individual wall uncertainties and statistical variation across repeated wall measurements used in the mass flow rate estimations. The overall mass flow rate result uncertainty is a combination of expanded sensor uncertainty ($k=2$) for all used sensors (CEN/TS 17660-1:2021, followed for gas sensors), detected concentration levels in relation to system detection limit ($<$ determination limit), number of repetitions for results, repeatability of detections in data collection, stability of weather conditions during data collection, consistency/variability of emission rates at sources and overlapping spread of emissions.

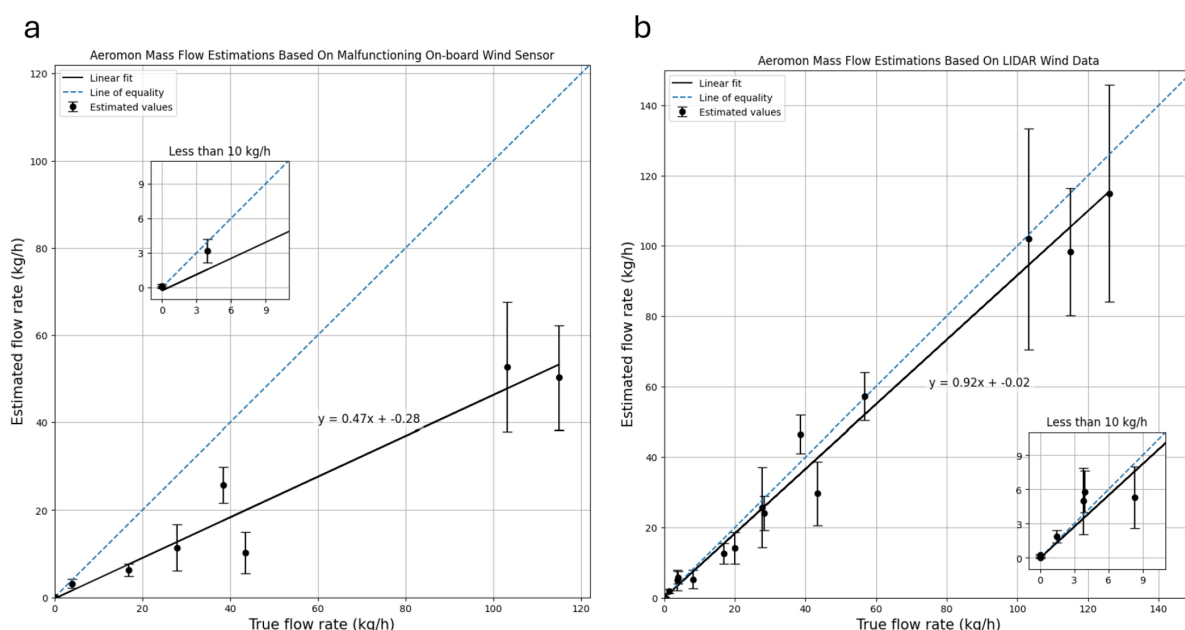


Figure 2—(a) Aeromon's originally reported mass flow estimations using the malfunctioning on-board wind data and (b) Aeromon's re-analysed mass flow estimations based on wind LIDAR data.

As part of the post-analysis, an evaluation on how repeated measurements affect test accuracy, expressed here as average error, was made. It is important to distinguish between the concepts of uncertainty and accuracy. Measurement uncertainty reflects the inherent imperfection of any measurement and is defined as a parameter that characterizes the range of values that could reasonably be attributed to the measured quantity. In contrast, accuracy refers to how closely a measurement aligns with the true value of the quantity being measured. In most real-world scenarios, accuracy cannot be directly assessed because the true value is unknown. However, controlled release studies provide a unique opportunity to evaluate accuracy, as the true release rate is precisely known. This analysis incorporated all individual wall measurements with LIDAR data, including partial walls that resulted from significant shifts in wind direction during some of the release tests. The findings from this expanded analysis are presented in Figure 3.

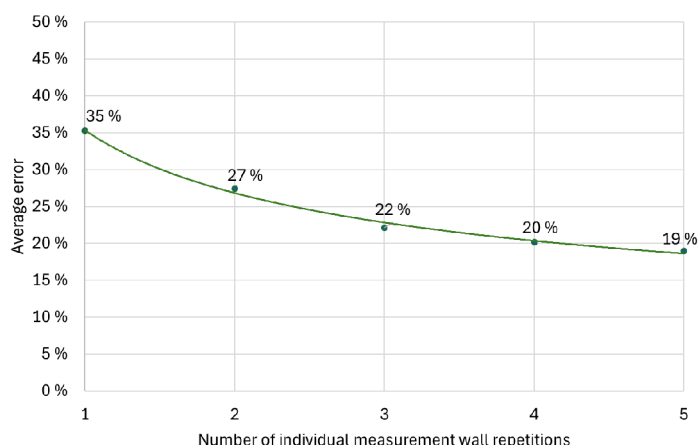


Figure 3—The accuracy dependency as a function of repetitions for all individual wall measurements throughout the campaign.

These results offer valuable insights into Aeromon's mass balance method's performance when wind data meets the minimum quality requirements. Although Aeromon's methodology was not originally designed for use with LIDAR-based wind data, the findings of result comparison and accuracy analysis provide a useful benchmark for assessing methodology's behavior under varying conditions and show clear potential in utilizing external wind data sources, assuming that they meet the data quality requirements. The comparative analysis during the TADI campaign revealed several factors contributing to Aeromon's underperformance in the field. Notably, LIDAR data does not inherently correct for unfavorable wind conditions — such as abrupt changes in prevailing wind direction — which can impact measurement accuracy regardless of the wind data source.

A comparison between the originally reported mass flow rates and the LIDAR-derived estimates indicated that weather conditions alone could not fully account for methodological performance during the release tests. As a result, Aeromon initiated a deeper investigation to identify the underlying causes of the suboptimal results. These observations allowed further refinement and optimization of Aeromon's measurement methodology and quality assurance procedures.

This paper summarizes the lessons learnt from the TADI campaign, outlines the investigation process where faulty wind sensor was identified, and describes the corrective actions implemented. It also discusses broader implications for methane emissions monitoring and emphasizes Aeromon's commitment to improving measurement precision and maintaining scientific rigor in industrial applications.

Methodology

In this chapter we present the key methodologies used in analysis and explain what all potential sources for the original analysis mismatch were investigated to explain the underestimated mass flow rates in the original reporting.

Mass flow rate quantification method – Mass Balance (MB)

Aeromon employed a drone-based MB measurement method to estimate methane mass flow rates during a single-blind controlled release campaign held in June 2024 at the TADI test centre. The MB method estimates mass flow by calculating the net flux of methane exiting an area that surrounds the emission source (Alfieri et al. 2010). Assuming a dominant horizontal wind direction, the method is simplified by measuring flux across two vertical planes: one upwind (capturing incoming flux) and one downwind (capturing outgoing flux). The calculation is based on real-time measurements of methane concentration $[CH_4]$ (g/m^3) and the component of wind speed perpendicular to the wall surface, $U_{\perp}$ (m/s). These values

are used to estimate the mass flow rate F (g/s) through a surface integral over the measurement walls. The used formula is:

$$F = \int_{x_1}^{x_2} \int_{z_1}^{z_2} \left([CH_4]_{downwind} - [CH_4]_{upwind} \right) U_{\perp} dx dz$$

The surface integral is approximated by summing over a discrete grid along the measurement walls, representing fence line measurements taken at a defined vertical resolution—2 meters in the case of the TADI campaign. Mass balance calculations were performed using Aeromon's ACS (Aeromon Cloud Service) software, which relies on a linear interpolation scheme for approximating the surface integral, [HS1] for analysis and reporting. However, a standalone local program also exists for special cases, such as non-standard (nonconforming) calculations. Additionally, the MB method is straightforward enough to allow for manual calculations where needed.

To rule out the possibility that the implementation of the MB method in software contributed to the suboptimal results, flux rates were calculated using all three alternative methods. The results were consistent, differing by no more than a tenth of a gram per hour, demonstrating the methods reliability and confirming the consistency of its implementation.

Weather conditions

Accurate methane mass balance (MB) analysis depends on clearly defined boundary conditions to ensure reliable measurements. Based on extensive field experience, Aeromon has developed strict operating procedures to manage the effects of weather and deliver consistent, scientifically valid results. The main conditions for suitable measurements are:

- Wind speed between 1–15 m/s (optimal range: 2–10 m/s)
- No significant rainfall
- Stable wind conditions with prevailing wind direction, typically with wind persistence above 0.9, as outlined in (EN 17628:2022)

To make sure that these requirements are met, Aeromon uses a dual-sensor setup: an on-board 3D wind sensor for localized wind data along the measurement lines, and a stationary weather station with metrologically traceable calibration for broader atmospheric monitoring. This layered approach normally ensures high accuracy and confidence in the measurement conditions and results.

However, during the TADI campaign, access to the stationary weather station was limited during the first half of the campaign due to logistical constraints. Because of the R&D nature of the campaign and a narrow measurement window, the decision was made to continue testing. This situation provided a valuable opportunity to assess how the MB method performs under suboptimal conditions—conditions that often closely resemble real-world challenges. Unfortunately, the absence of the secondary wind data source contributed to the on-board wind sensor malfunction going undetected.

In addition, independent observations showed that wind conditions during several test releases exceeded the defined measurement boundary conditions, making real-time assessments more difficult. For example, windsocks on-site visibly rotated more than 90 degrees during single wall measurement, indicating a lack of stable, prevailing wind—conditions that, under normal operations, would render the data nonconforming.

As part of the post-analysis, U (perpendicular wind component) values from the on-board sensor were plotted for each wall in every valid test. The goal was to assess how much time wind speeds fell below the acceptable lower limit of 1 m/s. Out of 44 measurement walls, only one had less than one minute (55 seconds) of wind speeds below this threshold, and only eleven others were below two minutes. Given that each wall was measured over approximately five to ten minutes, this analysis supports the conclusion that wind conditions were generally suboptimal for accurate measurement.

In the end, the combination of difficult weather conditions and the lack of a secondary source for real-time wind validation delayed the discovery of the on-board wind sensor malfunction. Since wind data is critical to the mass balance method, this issue directly affected the accuracy of the initial results. One key indicator of the problem was the frequent drops of the on-board sensor's perpendicular wind speed ($U_{\perp}$) into negative values, which helps explain the previously underestimated mass flow rates. An example of this behavior is also visible in Figure 4.

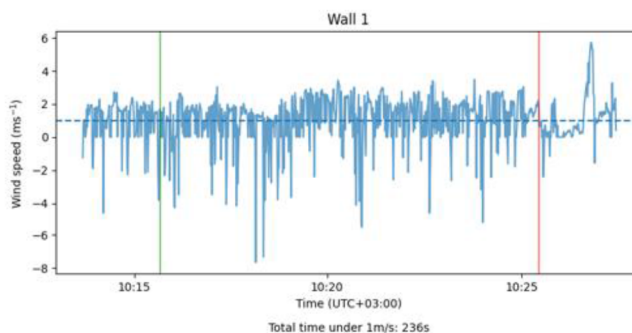


Figure 4—The $U_{\perp}$ values for an example wall from the TADI project, highlight that the wind conditions appeared to be unsuitable for measuring purposes. The green and red vertical lines mark the start and end of the active measurement periods, respectively.

The mass flow rates based on wind LIDAR data

The TADI site was equipped with a wind LIDAR sensor that operated throughout the campaign. At the request of Aeromon and other participating teams, measurement campaign organization provided access to the LIDAR data after the release of the actual flow rates, which helped reveal the actual wind conditions and supported flow rate analysis troubleshooting.

The shared LIDAR dataset includes wind speed and direction measurements from heights of 10 to 300 meters above ground level, with the lowest measurement levels at 10 m and 20 m — most relevant for this study. According to internal evaluations by TADI, the relative uncertainty of the wind speed measurements is estimated at 2.5–5%, making the dataset a reliable reference in the absence of a stationary weather station during the entire campaign.

Because Aeromon's measurements were taken at a vertical resolution of 2 meters above ground, LIDAR wind speeds at these lower heights were estimated. This is done using the wind profile power law [HS2] (Turner 1994), expressed by the following formula:

$$U_s(h) = \begin{cases} U_m \left(\frac{h}{h_m} \right)^p, & h > h_m \\ U_{mb}, & h \leq h_m \end{cases}$$

where U_s is the scaled wind speed at a desired altitude h , U_m is the measured wind speed at the measurement altitude h_m , and p is the wind profile exponent derived for each atmospheric stability class (A-F). Whilst the power law is limited in its efficacy at heights very close to the surface, this shortcoming was deemed acceptable in this circumstance, where resources were limited and the intended purpose of the calculations were purely illustrative. The wind LIDAR data has a temporal resolution of approximately 20 seconds. To align it with Aeromon's 1 Hz data resolution, a 60-second moving average was applied to the LIDAR data at a 1 Hz sampling rate.

An important indication of a potential issue with the on-board wind data came from a visual comparison between the on-board horizontal wind speed and the corresponding wind LIDAR data. The comparison revealed significantly larger fluctuations in the on-board measurements than in the LIDAR data. It's important to note that these two datasets may represent wind conditions at slightly different locations, though within the boundaries of the test site, a direct, point-by-point comparison is not meaningful. Instead, the

focus is on identifying overall trends over longer periods, such as one hour. Figure 5 illustrates the wind data for test #7 (17.6.2024) showing this pattern. Similar trends were observed across other tests during the campaign.

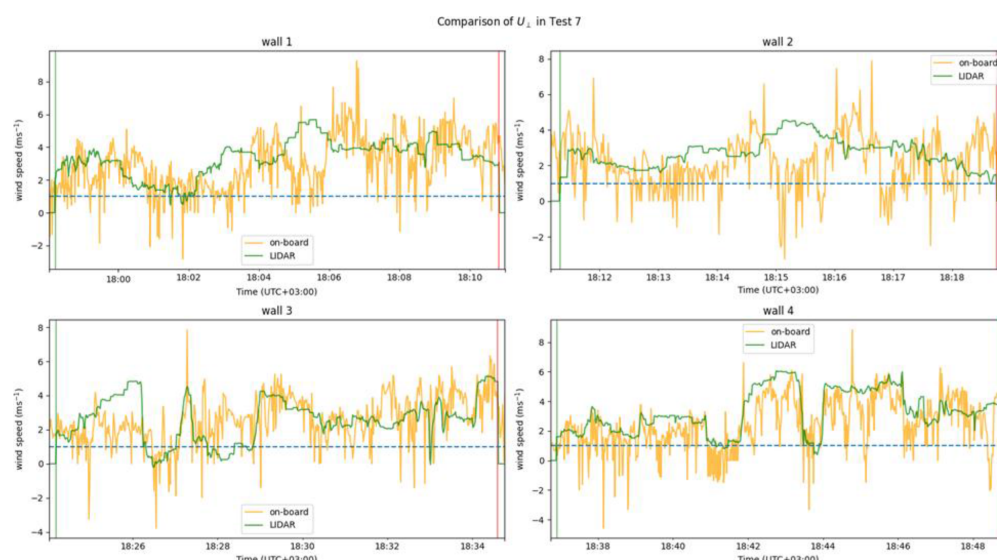


Figure 5—Comparison of the normal-to-wall wind speed measured by the on-board sensor and the corresponding values derived from wind LIDAR data (scaled from 10 m altitude) during test #7 of the TADI campaign, across four repetitions of downwind wall measurements.

Comparison between faulty and normally operating on-board wind sensors

In 2024, Aeromon formalized a verification procedure for testing the performance of on-board wind sensors used in field measurements. While now a standard part of Aeromon's calibration protocol, this procedure was still under development and revision during the TADI campaign. Since then, it has provided clear evidence that the on-board wind sensor used in the campaign was malfunctioning.

As part of a calibration evaluation conducted on September 19th, 2024, seven identical on-board wind sensors, including the one used at TADI, were installed alongside a stationary weather station (referred to as WXT) with metrologically traceable calibration. All sensors were mounted in an array (schematically shown in Figure 6) and exposed to identical wind conditions for one hour.

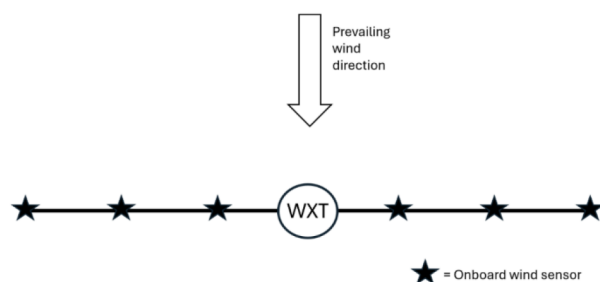


Figure 6—Schematic for the installation array for wind sensor reference test. The entire array length is approximately 2 meters.

The data from each on-board sensor was then compared to WXT reference using e.g. the following statistical metrics:

1. Mean Absolute Percentage Error (MAPE): where a MAPE of 0 indicates a perfect match with the WXT (Turner 1994).

- 2. Pearson correlation coefficient for absolute wind speeds: a value of 1 indicates perfect correlation with WXT, while 0 indicates no correlation (Wilks 2020)
- 3. The Mean Absolute Circular Error (MACE) is a metric used to assess the accuracy of circular data comparisons, particularly for measurements like wind direction or any data that wraps around in a circular manner (e.g., angles between 0° and 360°) (Bao et al. 2010). Unlike linear data, circular data has the challenge of being periodic, meaning the values wrap around 360° is effectively equivalent to 0°. This makes traditional error metrics (like the Mean Absolute Error, MAE) unsuitable for circular data, as they don't account for the cyclic nature of the values.

The results, summarized in Table 1, show a clear mismatch between sensor 7 (used during TADI) and the WXT reference. Moreover, sensor 7 also failed to align with the other six on-board sensors, which were in good agreement with each other. Notably, sensor 7 recorded instances of negative absolute wind speeds, which are physically implausible.

Table 1—Statistical comparison metrics between the on-board wind sensors and the calibrated reference weather station (WXT) from calibration evaluation conducted on September 19th, 2024.

	MAPE wind speed (%)	Pearson cc wind speed (A.U.)	Mean MACE (°)
WXT	0	1	0
sensor 1	26.4	0.89	25.2
sensor 2	23.0	0.91	21.7
sensor 3	22.9	0.91	21.9
sensor 4	21.8	0.91	20.4
sensor 5	23.4	0.90	22.3
sensor 6	24.2	0.92	25.4
sensor 7	83.2	0.34	30.3

Even after filtering out these invalid points, sensor 7 remained an outlier. While sensors 1–6 showed close agreement, sensor 7 exhibited much higher variability. Figure 7 illustrates this discrepancy, comparing sensor 7's output to the point-wise average of the other six sensors.

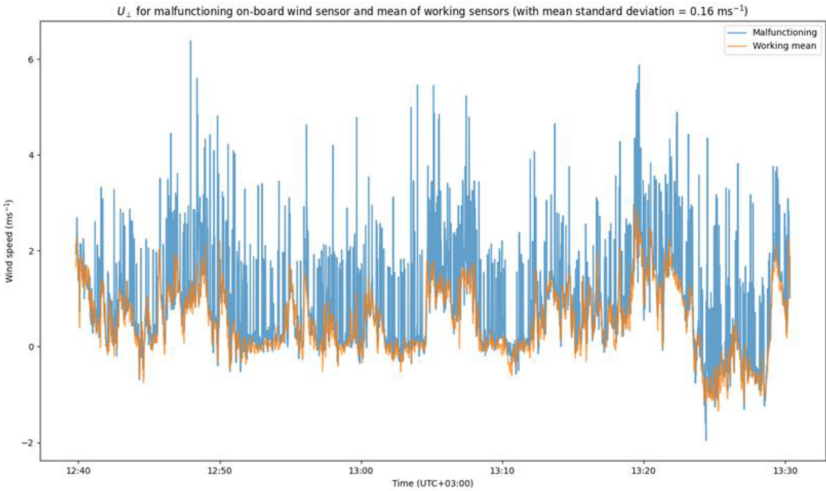


Figure 7—During the on-board wind sensor test, the malfunctioning sensor showed significantly greater variability compared to the other sensors, which had an average standard deviation of just 0.16 m/s among themselves.

After identifying the malfunctioning on-board wind sensor, immediate steps were taken to assess and mitigate any potential impact. The sensor was promptly removed from service, and all related measurements were revalidated to ensure data accuracy and integrity. In each case, a consistent trend of lower-than-expected wind speeds was observed, underscoring the importance of timely corrective measures.

To better understand and potentially correct the sensor's malfunction, its behavior was analyzed relative to the functioning sensors in the comparison test. When the malfunctioning sensor's readings were plotted against the average values from the operational on-board sensors in a scatterplot (see Figure 8), distinct linear bands emerged. This pattern suggests that the faulty sensor was primarily affected by a consistent offset in its measurements.

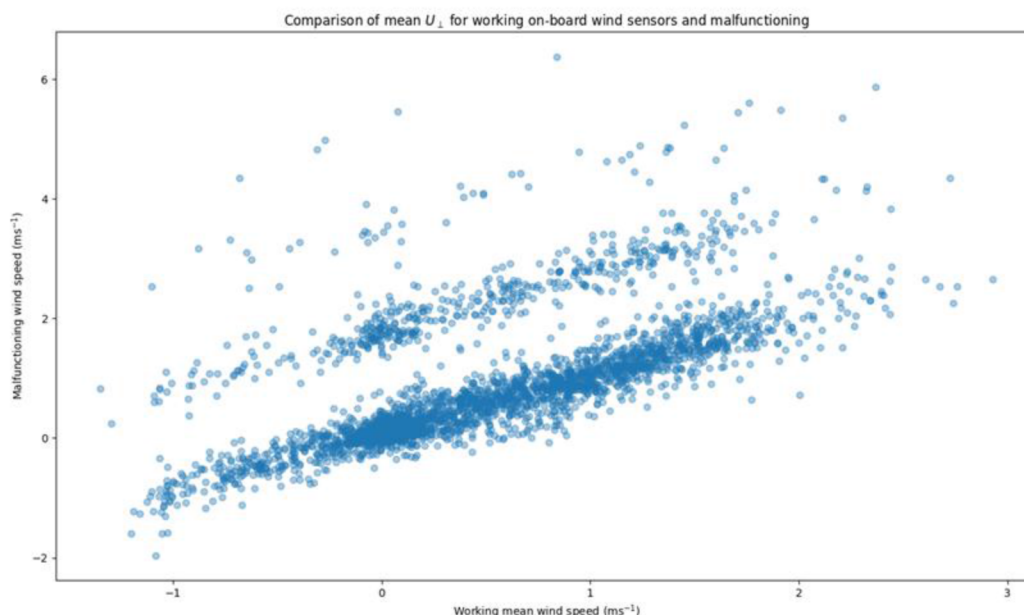


Figure 8—Plotting the malfunctioning on-board wind sensor data against the average of the functioning sensors reveals a series of distinct linear bands, suggesting that a linear correction to the data may be possible.

Attempts to classify the data beyond visual inspection resulted in successful grouping of approximately 95% of the readings, with the remaining 5% considered noise. This indicates that, under ideal conditions where a secondary reference sensor is available, the offset could be corrected for the majority of the data—leading to improved accuracy. However, due to the structure of the groupings, it was not possible to define a reliable method for adjusting the offset on a point-by-point basis.

As a result, the data from the malfunctioning sensor was considered unreliable for this project. To uphold Aeromon's commitment to providing accurate and dependable measurements, alternative wind data sources were identified and used for flux rate calculations, ensuring the integrity of the final results.

Conclusion

The TADI campaign served as a crucial opportunity to evaluate the reliability of Aeromon's methane mass balance (MB) measurement methodology under controlled conditions. The discovery that a malfunctioning on-board wind sensor was the primary cause of discrepancies in reported mass flow rates underscores the importance of systematic data validation and of incorporating redundancy into measurement campaigns. A subsequent re-analysis using wind LIDAR data confirmed that, when accurate wind input is available, the MB method produces results that closely match actual release rates. To improve future measurement accuracy, Aeromon carefully reviewed the factors that delayed the detection of this issue and identified the following key contributors:

1. Limited redundancy due to logistical constraints: The stationary weather station was unavailable during the first two days of the campaign, preventing secondary validation of wind conditions during this critical period. This emphasized the need for layered verification systems, using both stationary and on-board wind measurements used in normal operations, as described in current accredited methodologies for (ISO 17025:2017).
2. Challenging weather conditions: Unstable weather during the campaign masked the effects of the sensor malfunction. High wind variability made it difficult for field operators to identify faults in real time, especially in the absence of a clear reason to suspect sensor issues. In normal production measurements the accredited methodology clearly describes acceptable boundary wind conditions for data collection; had these measurements been for normal production--and not research--purposes, the project would have been rescheduled for more sufficient weather conditions.
3. Pending implementation of ISO 17025-compliant quality control and assurance protocols[HS3]: Aeromon had already recognized the need for systematic pre-field validation of on-board sensors, in line with accreditation requirements for (ISO 17025:2017). However, the full implementation of this process was not completed at the time of the measurements. Following the TADI campaign, Aeromon expedited the rollout of these procedures to strengthen sensor reliability. The quality assurance protocols for on-board wind sensors were designed to i) reliably identify malfunctions similar to the one encountered in the TADI measurements and ii) to extend the traceable calibration of wind speed and direction measurements to the on-board wind sensors. This incident, being the first recorded on-board wind sensor failure in the field, provided valuable lessons that have since informed and improved validation protocols.

After Aeromon had reference tested the entire wind sensor population against the metrologically traceable weather station, different options for the acceptance criteria were analyzed thoroughly. From all the tested statistical tools, the Pearson correlation coefficient for wind speed was chosen as the key metrics for acceptance in reference testing analysis of the on-board wind sensors. The selection was based on consistent and clear outcome between working and faulty sensor units in all tested wind conditions for the entire sensor population. The Pearson correlation coefficient is a statistical measure used to estimate the strength and direction of the linear relationship between two continuous variables based on a sample of data. The Pearson correlation coefficient is calculated as follows:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

where $\bar{x}$ and $\bar{y}$ are the means of variables x and y . r is between -1 (negative correlation) and 1 (positive correlation). If $r = 0$, there is no correlation. A Pearson correlation coefficient of 0.7 and above is typically considered a strong positive correlation. A Pearson correlation coefficient of 0.5 indicates a moderate positive correlation. A correlation value between 0 and 0.4 is considered a weak correlation (de Myttenaere et al. 2016). This reference testing is performed on all on-board wind sensors prior to being released to production and regularly thereafter. This reference testing also enables extending the metrologically traceable calibration of the weather stations to on-board wind sensors within Aeromon's test laboratory accreditation.

As the TADI campaign was conducted as part of a research and development effort, even the data affected by poor weather offered useful insights for methodological refinement. Under standard operational conditions, such measurement anomalies would have triggered a remeasurement in accordance with Aeromon's service procedures. Despite the challenges, the campaign yielded important lessons that have led to significant improvements in Aeromon's quality control and assurance practices within the scope of its ISO 17025 test laboratory accreditation and sensor validation protocols, as well as its strategies for sources

of alternative wind data. These advancements will enhance the accuracy and reliability of future airborne methane monitoring campaigns.

Discussion

A comprehensive analysis of the on-board wind sensors has confirmed that the mass flow estimates from the TADI campaign were significantly affected by a malfunctioning wind sensor. This conclusion is supported not only by direct analysis of the faulty sensor itself but also by comparisons with an alternative wind dataset from a LIDAR system. Visual comparisons between multiple on-board sensors and the LIDAR data clearly show that the readings from the malfunctioning sensor deviated substantially, significantly influencing the MB (mass balance) analysis.

The sensor malfunction was the primary reason the initial reported values failed to meet Aeromon's internal quality standards. In response, Aeromon has implemented a series of improvements, including a standardized reference test protocol for on-board wind sensors, enhanced monitoring of sensor performance, and the integration of alternative wind data sources such as LIDAR. These steps collectively improve the accuracy and reliability of methane quantification, particularly under challenging wind conditions or when sensor performance is uncertain.

The insights gained from this investigation have strengthened Aeromon's quality management systems, leading to more rigorous sensor validation processes and greater data integrity in measurement campaigns. A new testing protocol for on-board sensors is now integrated into service production, and the comparison studies using LIDAR have deepened understanding of how critical accurate wind measurements are to mass flow calculations—knowledge that will continue to shape Aeromon's methodology going forward. As part of its continued commitment to research and development, Aeromon has used the findings from the TADI campaign to refine its quality assurance protocols, ultimately delivering more robust and scientifically grounded measurement services. The campaign raised important questions that have proven invaluable to Aeromon's mission of delivering accurate, traceable, and reliable emissions data.

Following the campaign, Aeromon achieved formal laboratory accreditation (FINAS 2024), incorporating the quality assurance and traceability improvements identified during this process into its certified quality system. These enhancements have further improved consistency, reliability, and accuracy of measurement results.

Beyond addressing immediate concerns, this study contributes to the ongoing advancement of emissions measurement practices. It underscores the importance of using multiple wind data sources, ensuring redundancy in field sensors, and applying flexible data processing techniques to handle variable environmental conditions. By applying these lessons, Aeromon continues to refine its methane mass flow estimation approaches, enhancing both precision and resilience across a range of measurement scenarios and methodologies. This iterative improvement process ensures that emissions monitoring remains rigorous, traceable, and aligned with the highest technical standards for quantification and reporting.

References

- Alfieri, S., Amato, U., Carfora, M. F. et al 2010. Quantifying trace gas emissions from composite landscapes: A mass-budget approach with aircraft measurements. *Atmospheric Environment* **44**: 1866–1876. <https://doi.org/10.1016/j.atmosenv.2010.02.026>.
- Bao, L., Gneiting, T., Gritti, E. P. et al 2010. Bias Correction and Bayesian Model Averaging for Ensemble Forecasts of Surface Wind Direction. *Monthly Weather Review* **138**: 1811–1821, <https://doi.org/10.1175/2009MWR3138.1>.
- CEN/TS 17660-1:2021, Air quality - Performance evaluation of air quality sensor systems Gaseous pollutants in ambient air. 2021.
- de Myttenaere, A., Golden, B., Le Grand B. et al 2016. Mean Absolute Percentage Error for regression models. *Neurocomputing* **192**: 38–48. <https://doi.org/10.1016/j.neucom.2015.12.114>.

- EN 17628:2022, Fugitive and diffuse emissions of common concern to industry sectors - Standard method to determine diffuse emissions of volatile organic compounds into the atmosphere. 2022. Europe: European Committee for Standardization (CEN).
- FINAS. 2024. Aeromon Oy Accredited Testing Laboratory. Final report, Finnish Accreditation Service (FINAS), Helsinki, Finland. https://www.finas.fi/Documents/T362_A01_2024.pdf
- Hamedani Raja, S., Bullock, T., Laurikainen, M. et al 2025. Assessing anomalies in methane mass flow estimation: Investigating findings and lessons for enhanced monitoring. *Zenodo*. <https://doi.org/10.5281/zenodo.14882795>.
- ISO 17025:2017, General requirements for the competence of testing and calibration laboratories. 2017. Geneva, Switzerland: International Organization for Standardization committee on conformity assessment (ISO/CASCO).
- Turner, D. B. 1994. Workbook of Atmospheric Dispersion Estimates, second edition. Boca Raton, Florida, USA: CRC Press.
- Wilks, D. S. 2020. Statistical Methods in the Atmospheric Sciences, fourth edition. Amsterdam, Netherlands: Elsevier.